Bilateral Photoplethysmography Analysis for Peripheral Arterial Stenosis Screening With a Fractional-Order Integrator and Info-Gap Decision-Making

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***Abstract*—Peripheral arterial disease and atherosclerosis are common complications in patients with type 2 diabetes or with both diabetes and end-stage renal disease. Lower-limb peripheral arterial disease and hemodialysis (HD) vascular access stenosis are highly prevalent in HD patients; in particular, progressively narrowed vascular access and suboptimal dialysis blood flows are the major issues. In this paper, we propose a fractional-order integrator (FOI) for indicating the differences in the rise-timing and amplitudes of bilateral photoplethysmography (PPG) signals with the aim of improving HD healthcare. The area under the systolic peak ratio was used as an index to separate the patients without arterial stenosis from those with arterial stenosis; subsequently, info-gap decision making was conducted to evaluate risk levels in reference to the patients' health records and the degrees of stenosis. The experimental results indicated that in comparison with 1) the resistive index; 2) bilateral timing parameters; and 3) the hybrid intelligent method, our proposed screening model was more efficient in preventing complications of peripheral arterial stenosis and was easily implemented in an embedded system.**

***Index Terms*—Hemodialysis, fractional-order integrator, area under the systolic peak ratio, info-gap decision-making.**

List of Abbreviations and Nomenclatures

HD	Hemodialysis
PAD	Peripheral Arterial Disease
CVD	Cardiovascular Disease
ESRD	End-Stage Renal Disease
PPG	Photoplethysmography
FOI	Fractional-Order Integrator
FOD	Fractional-Order Derivative
I-G	Info-Gap
AUSP	Area Under the Systolic Peak
AUSPR	Area Under the Systolic Peak Ratio
RT	Rise Time
PTT	Pulse-Transit Time
RI	Reflection Index
SI	Stiffness Index
CF	Certainty Factor
ANN	Artificial Neural Network

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I. Introduction

IN TAIWAN, cardiovascular diseases (CVDs) are major risk factors for morbidity and mortality in patients with diabetes or with both diabetes and end-stage renal disease (ESRD). Patients who suffer from progressive atherosclerosis can develop arterial stenosis, vascular diseases, or CVDs, which can include complications such as hypertension, hyperlipidemia, dyslipidemia, and diabetes mellitus. Diabetes mellitus is the most common cause of chronic kidney failure, whereas endothelial dysfunction is a primary contributing cause of atherosclerosis. In vascular access studies, pathological findings include vascular smooth muscle proliferation, endothelial cell activation, and extracellular matrix accumulation [1], [2] in the atherosclerosis of peripheral arteries, coronary arteries, and the aorta. Lower-limb PAD is highly prevalent in patients with type 2 diabetes mellitus; it can lead to narrowed vascular access, leg pain, and/or foot gangrene that requires amputation [3]. In addition, intradialysis complications, such as venous side stenosis or stenoses near the graft-to-vein site, frequently occur with HD, and these can reduce dialysis blood flow [4]. Adequate blood flow is considered as >600 ml/min for arteriovenous fistulas and >400–500 ml/min for arteriovenous grafts. In this study, we propose a prognostic detection model consisting of bilateral PPG and an inference method that can be used to screen for arterial stenosis in patients with diabetes and ESRD.

PPG signals, which are used in clinical examinations, carry information about cardiovascular function, heart rate, and

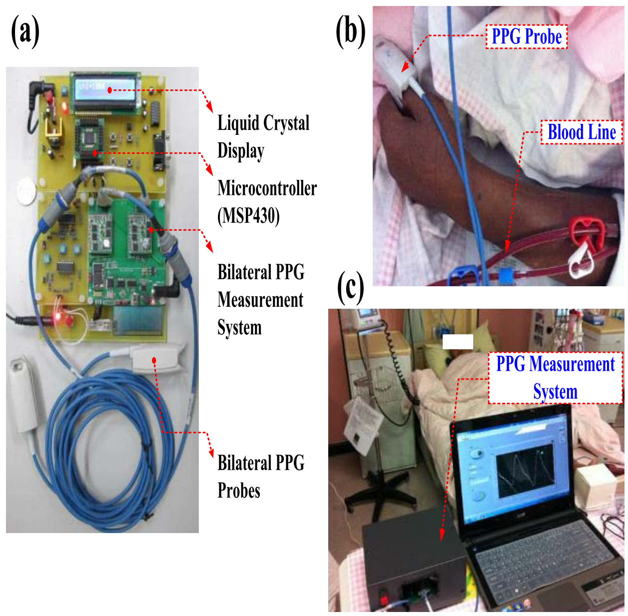

Fig. 1. PPG optical measurement technique. (a) Bilateral PPG measurement system with two probes. (b) PPG probe and blood line. (c) Practical bilateral PPG measurements.

arterial/venous blood oxygen saturation [5]–[9]. Optical measurement techniques (red and infrared wavelengths) allow for comfortable and non-invasive monitoring of bio-signals and the placement of reflecting or transmission modes on any skin surface. Therefore, optical probes and measurement hardware are easily integrated into mobile or wearable devices [5], [6], as shown in Figure 1. With bilateral or synchronous multi-site measurements, previous studies have shown significant bilateral PPG differences in CVDs, vascular diseases, and hypertension [10]–[13]. An earlobe, thumb, index finger or big-toe clip incorporating a light-emitting diode and a photodetector enables the detection of vasoconstriction and vasodilatation in vessels <2.5 mm in diameter. Continuous measurement and temporal analysis can be used to detect changes in blood volume and blood circulations at different body sites.

Both the time-domain [7], [8], [10], [11] and frequency-domain [5], [9], [12] methods have been applied to analyze PPG signals for hand and leg arterial stenosis screening. Time-domain methods have identified differences in bilateral timing parameters and pulse waveforms in both upper and lower limbs that can separate diabetic and non-diabetic subjects. As the severity of stenosis evolves, the dexter-to-sinister dissimilarity in transit time and pulse amplitudes increases significantly. A quantitative method based on fractional-order differentiation formulations [9], [14] is also used to represent the differences in the transit time and amplitudes of bilateral PPG signals. However, limitations related to numerical computation, memory, and sampling data requirements remain an issue for utilizing mobile or wearable devices in real-time analysis. Frequency-domain methods [9], [10], such as Fourier and wavelet transformation, provide bilateral frequency spectra with similarity characteristics in higher frequency components for healthy subjects and bilateral asymmetry with lower-frequency components in diabetic patients.

The aforementioned methods show potential for use in the screening of arterial stenosis. For real-time analysis and computations, discrete fractional order integrators (FOIs) [16]–[19] with finite computations, short-memory requirements, and finite power series are employed to calculate the bilateral pulses' area under the systolic peak (AUSP) [20]. The AUSP ratio (AUSPR) is validated to identify the patients with and without PAD. Subsequently, info-gap (I-G) [21]–[23] decision-making utilizes a screening model to automatically detect diabetic and HD patients with PAD. The remainder of this paper is organized as follows: Section 2 describes the PPG measurement system and the definition of the AUSPR; Section 3 addresses the methodology of the study, including the discrete FOI, I-G decision-making, and enrolment of subjects; and Sections 4 and 5 present the experimental results and conclusions, respectively, which demonstrate the efficiency of the proposed screening model.

II. PPG Measurement System

PPG is a non-invasive optical measurement technique that provides an estimation of blood variations and blood transport via the measurement of dynamic attenuation of red or infrared light, either bilaterally or unilaterally, at different body sites such as the earlobes, index fingers, thumbs, and big toes [6]–[8]. A PPG waveform consists of AC and DC components that carry physiological information about heart rate, cardiovascular regulation, thermoregulation, respiration, blood changes in peripheral arteries, and endothelial function. PPG can be used to evaluate cardiovascular risk factors, atherosclerosis, arterial stenosis, arterial properties, and hypertension [10]–[13]. Dual-channel PPG (pulse oximeter) signals with different wavelengths (red or infrared light) can be used to estimate instantaneous venous oxygen saturation, which provides valuable clinical information toward detection of the early phases of shock [15].

A PPG pulse is attributable to variation in blood volume caused by the pressure pulse of each cardiac cycle. The systolic peak is a result of pressure wave from the left ventricle to the periphery of human body, as seen in Figure 2. The rising edge (systolic region) of each PPG pulse is an interval revealing vascular compliance. Some indexes, such as the reflection index (RI), augmentation index, stiffness index (SI), as well as the area under the systolic peak and timing parameters are used to evaluate arterial stiffness, arterial compliance, and vascular health [10], [11], [24]–[26]. Age is also an important factor that affects the contour of a PPG signal; for example, arterial compliance degrades with age due to arterial elasticity. For PPG morphology in older and unhealthy subjects, the dicrotic notch and diastolic peak may disappear [24]–[26]. The RI and SI depend on the diastolic peak; thus, these parameters cannot be used for subjects of all ages. Timing parameters, such as rise time (RT) and pulse-transit times (PTTs) [10], [11], show that for bilateral differences, $\Delta\mathrm{RT} = |\mathrm{RT}_R - \mathrm{RT}_L|$ and $\Delta\mathrm{PTT} = |\mathrm{PTT}_R - \mathrm{PTT}_L|$ (dexter-to-sinister) as vascular disease gradually evolves; this is highly correlated with the evaluation of vascular diseases and arterial stenoses.

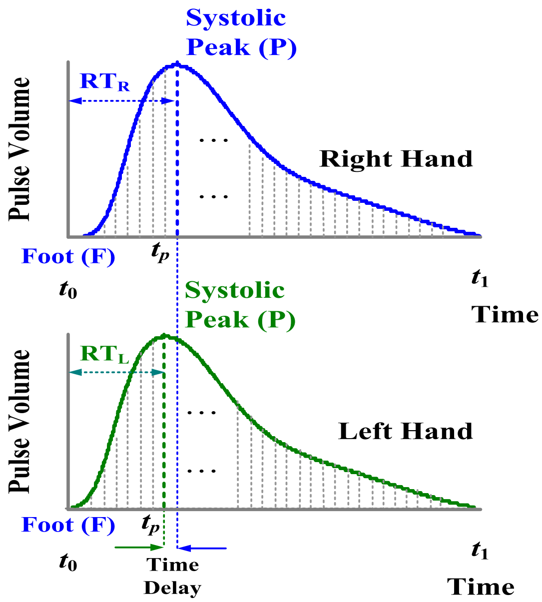

Fig. 2. Bilateral PPG waveform, including timing parameter, rise time, and delay time.

However, transit timing difference methods need an extra R peak from an electrocardiogram as a timing reference. As seen in Figure 2, when vascular compliance degrades or vascular resistance increases, delayed transit time extracted from bilateral PPG is an indicator for PAD screening. In addition, changes in amplitudes and pulse shapes, i.e., the AUSP, can also be calculated to evaluate the bilateral differences from the pulse foot (F) to the systolic peak (P) [20], where symbols, "F" and "P", are indicated in Figure 2. The AUSP provides a simple non-invasive index to indicate the changes in the elastic properties of blood vessels. Hence, AUSPR is used to evaluate bilateral similarity and is defined as follows:

$$AUSPR = \frac{AUSP_R}{AUSP_L} = \frac{\int_{t_0}^{t_P} PPG_{R,FP}(t)dt}{\int_{t_0}^{t_P} PPG_{L,FP}(t)dt} \quad (1)$$

$$\Rightarrow AUSPR = \begin{cases} > 1, & AUSP_R > AUSP_L \\ \approx 1, & AUSP_R \approx AUSP_L \\ < 1, & AUSP_R < AUSP_L \end{cases} \quad (2)$$

where $AUSP_R$ and $AUSP_L$ are the dexter and sinister areas under the systolic peak, respectively; $PPG_{R,FP}(t)$ and $PPG_{L,FP}(t)$ are the dexter and sinister PPG signals from the pulse foot (F) to systolic peak (P), respectively; and $[t_0, t_P]$ is the time interval. *AUSPR*s>1 or <1 represent bilateral differences in peripheral arterial stenosis screening.

III. Methodology

A. Discrete FOI

Traditional integer derivatives or integral computations use the integer-order differential operators with Tustin transforms and trapezoidal integration methods for geometrical interpretations. Both the time and frequency domain behaviors of some processes do not fit the exponential evolution in the time domain or integer order slopes in their responses, e.g., the charging and discharging of capacitors, fluid flow in a porous media, and heat conduction in a semi-infinite slab. These processes can be more accurately modeled using fractional differ-integrals [16]. In contrast, fractional order calculus is calculated with respect to time and can be approximated using high-order polynomials with integer order differ-integral operators [17]–[19]; thus, it is employed to describe dynamic phenomena, both in the time and frequency domain behaviors. It can be used to express the finite power series/series expansion of non-linear functions as a summation of weighted coefficients.

For signal processing, the FOI employs all non-integer and irrational numbers to deal with signals, and the summation is computed using the ratios of the gamma function, $\Gamma(\alpha)$, which incorporates the number of sampling data points in the time interval $[t_0, t_1]$, and the fractional order parameters, α [19]. Therefore, the FOI has a broad application range for the analysis of time-varying signals as a function of time, $f(t)$, which includes fluid flows, heat conduction, and bio-signals. For the purpose of discrete computation, an FOI using the Grünwald–Letnikov (G–L) definition is given by [18] as follows:

$$D^{-\alpha} f(t) \approx \lim_{\Delta\tau\to 0} (\Delta\tau)^{\alpha} \sum_{i=0}^{n-1} \left|(-1)^i \binom{-\alpha}{i} f(t - i\cdot\Delta\tau)\right| \quad (3)$$

where $0 < \alpha < 1$ for FOI process, $\alpha \in$ R; $-1 < \alpha < 0$ for fractional-order derivative (FOD) process; integer, $\alpha = +1$, for integer-order differentiation and $\alpha = -1$ for integration process; n is the number of sampling point; and the binomial coefficients are given as

$$\binom{\alpha}{0} = 1, \quad \binom{-\alpha}{i} = \frac{-\alpha(-\alpha-1)\cdots(-\alpha-i+1)}{i!} = \frac{\Gamma(-\alpha-1)}{\Gamma(i+1)\Gamma(-\alpha-i+1)} \quad (4)$$

$$\text{time step:} \Delta\tau = (\frac{t_1 - t_0}{n}) \quad (5)$$

For bilateral PPG signals, the AUSPR was calculated and bounded in each time interval $[t_0, t_p]$. Hence, the AUSPR as equation (1) can be modified as a discrete fractional-order formulation as follows:

$$AUSPR \approx \frac{\lim_{\Delta\tau_R\to 0} (\Delta\tau_R)^{\alpha} \sum_{i=0}^{n_R-1} \left|(-1)^i \binom{-\alpha}{i} PPG_{R,FP}[i]\right|}{\lim_{\Delta\tau_L\to 0} (\Delta\tau_L)^{\alpha} \sum_{i=0}^{n_L-1} \left|(-1)^i \binom{-\alpha}{i} PPG_{L,FP}[i]\right|} \quad (6)$$

time step:

$$\Delta\tau_R = \frac{t_{R,FP} - t_0}{n_R}) \text{ and } \Delta\tau_L = (\frac{t_{L,FP} - t_0}{n_L}) \quad (7)$$

where the discrete dexter-PPG signal is referred to as $PPG_{R,FP}[i]$ in time interval $[t_0, t_{R,FP}]$, $i \in [0, n_R - 1]$ and the discrete sinister-PPG signal is referred to as $PPG_{L,FP}[i]$ in time interval $[t_0, t_{L,FP}]$, $i \in [0, n_L - 1]$. The bilateral PPG signals can be defined as follows:

$$PPG_{R,PF}[i] = \frac{PPG_R[i]}{PPG_{R,\max}}, \quad i \in [0, n_R - 1] \tag{8}$$

$$PPG_{L,PF}[i] = \frac{PPG_L[i]}{PPG_{L,\max}}, \quad i \in [0, n_L - 1] \tag{9}$$

where $PPG_{R,max}$ and $PPG_{L,max}$ are the systolic peak, and the initial conditions are $PPG_R[0] = PPG_L[0] = 0$. The AUSPR is a function of the variable fractional order, $0 < \alpha < 1$, and it has a broad range for extraction of the characteristics of time-varying PPG signals. Numerical evaluation with finite computations, short-memory requirements, and finite power series are advantages for computer programming, summation operations, and numerical regression in real-time analysis.

For bilateral PPG measurements, two-channel signals were synchronously captured with a data acquisition controller (1kHz sampling rate) from the PPG probes; these were passed to an embedded system (MSP430), which provided an analog–to–digital data transfer to a tablet PC, as seen in Figure 1(a). Work related pulse transit time used 1 kHz sampling rate to obtain sampling data in the rising edge. For further signal analysis and discrete FOI computations, the statistical data from 35 subjects were used to design the screening model. The patients enrolled in the study consisted of normal subjects, HD patients, and diabetic patients. These were selected from the Institutional Review Board (IRB) of Kaohsiung Veterans General Hospital (KVGH), Tainan Branch, under contract number: VGHKS13-CT12-11. Based on subjects with and without vascular diseases, the subjects were divided into four groups: a) healthy subjects (n=8); (b) diabetic subjects with low-grade and high-grade PAD (n=3); (c) HD patients without PAD (n=7); and (d) HD patients with PAD (n=7).

The data acquisition controller located each pulse foot, "F", to systolic peak, "P", to acquire sampling data within the F-P interval, as the sampling point versus the normalized amplitude, as seen in Figures 3(a) and 3(b). These patterns represented the bilateral PPGs of a normal subject, and they were in contrast to the dexter-to-sinister asymmetry in the amplitude, RTs, and AUSP of subjects with PAD. For feature extraction, an FOI with the fractional order, α=0.1–0.4 or α>0.4, was calculated using equations (3) to (5), as shown in Figures 3(c) and 3(d). Distinguishable features with a fractional order α=0.1–0.2 were observed. Hence, α=0.1 was selected to separate the distinction used in this study. For the four groups of subjects, fractional order AUSPRs were calculated using equations (6) to (9), as shown in Figure 4. According to the bilateral timing differences, ΔRT, these preliminary results separated normal subjects from subjects with PAD in three ranges with the boundaries <0.95, 0.95–1.05, and >1.05. ΔRT gradually increased as the AUSPRs were removed from the normal ranges. Three preliminarily boundaries were confirmed to include PAD in ranges <0.95 and >1.05; otherwise, the normal condition was confirmed to be in the range 0.95–1.05.

B. I-G Decision-Making

An I-G distribution model can be used to represent a wide variety of prior information. It is a nested family of the set

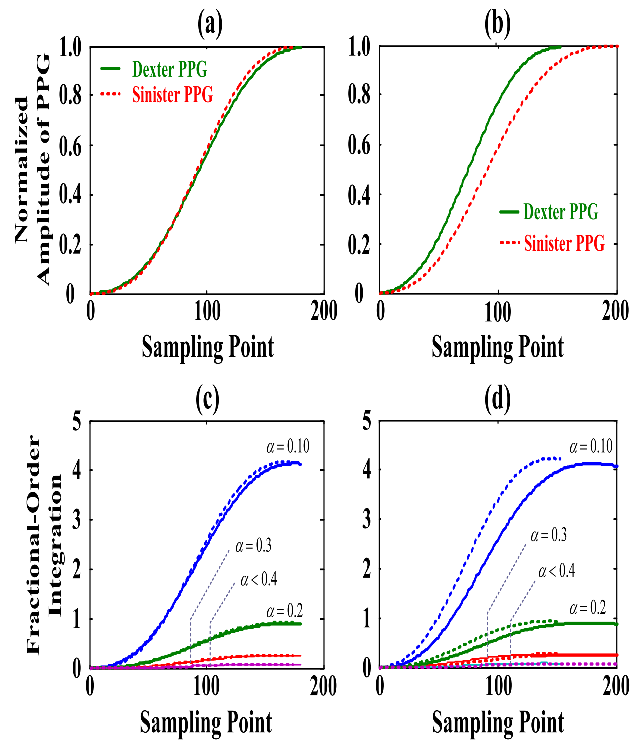

Fig. 3. PPG features. (a) and (b) PPG pattern in dexter-to-sinister interval from normal subject and subject with PAD, (c) and (d) FOI computations with fractional order, $\alpha = 0.10 \sim> 0.40$.

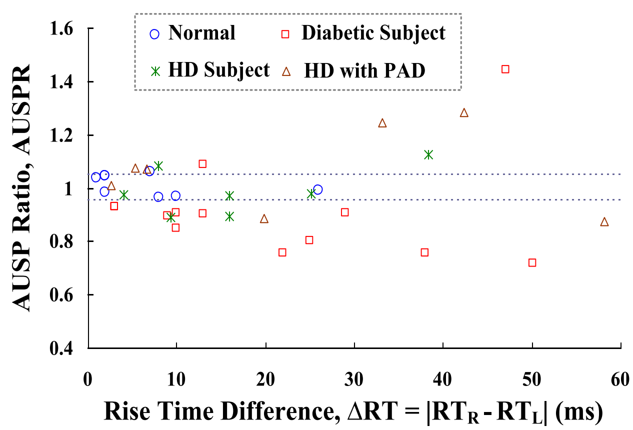

Fig. 4. AUSP ratios versus rise time difference for 4 groups of subjects.

of all univariate distribution functions, such as the probability distribution and the function of a relationship between two or more key parameters [21]–[23]. A common model can be represented as follows:

$$U(\beta, u') = \{u : |u - u'| \leq \beta\}, \quad \beta \geq 0 \tag{10}$$

where u' is the best estimate of an uncertain function u, while β is the fractional error from the estimation. The set $U(\beta, u')$ contains all functions u, while the fractional errors from the best functions u' are not greater than β.

As seen in Figure 4, bilateral timing differences, ΔRT, gradually increased with increasing AUSPRs for $AUSP_R > AUSP_L$, and AUSPRs decreased for $AUSP_R < AUSP_L$, respectively.

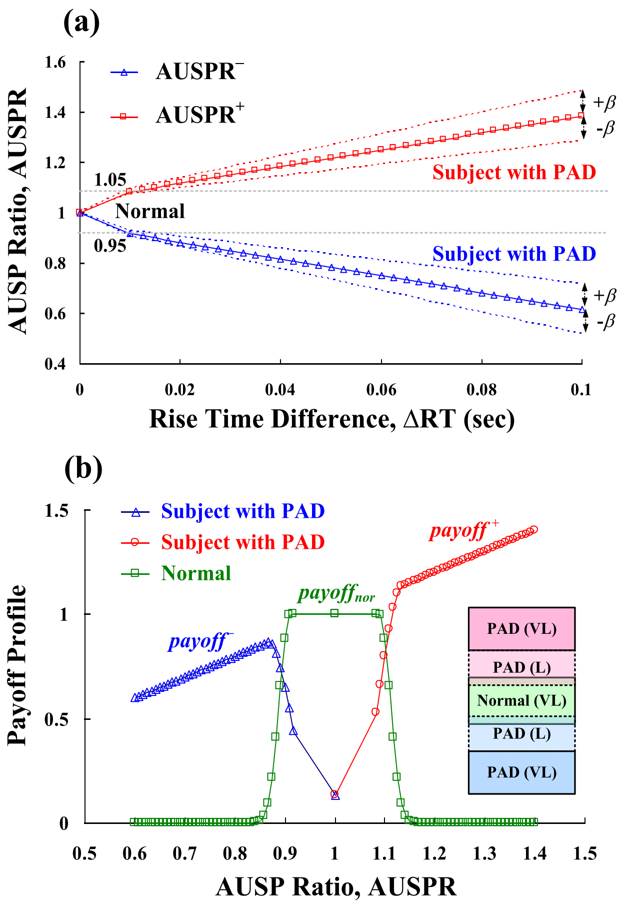

Fig. 5. Feature profiles. (a) The piecewise approximation functions between the AUSPR and the ΔRT. (b) The payoff profiles versus the AUSPRs for normal and subject with PAD.

The I-G distribution model can be used to represent the piecewise linear functions between the AUSPR and the ΔRT, as shown in Figure 5(a). We have two approximation functions, u^{+}=AUSPR^{+} and u^{-}=AUSPR^{-}, within the maximum (max) and minimum (min) interval, [$AUSPR_{min}$, $AUSPR_{max}$] and [ΔRT_{min}, ΔRT_{max}], as follows:

$$u^{+} = \begin{cases} (\dfrac{AUSPR_{\min}-1}{\Delta RT_{\min}})\Delta RT + 1.00, \\ \qquad\qquad 0 \le \Delta RT \le \Delta RT_{\min} \\ (\dfrac{AUSPR_{\max}-AUSPR_{\min}}{\Delta RT_{\max}-\Delta RT_{\min}})\Delta RT + 1.05, \\ \qquad\qquad \Delta RT > \Delta RT_{\min} \end{cases}$$

$$= \begin{cases} K_1^{+}\Delta RT + 1.00, & 0 \le \Delta RT \le \Delta RT_{\min} \\ K_2^{+}\Delta RT + 1.05, & \Delta RT > \Delta RT_{\min} \end{cases} \tag{11}$$

$$u^{-} = \begin{cases} -K_1^{+}\Delta RT + 1.00, & 0 \le \Delta RT \le \Delta RT_{\min} \\ -K_2^{+}\Delta RT + 0.95 & \Delta RT > \Delta RT_{\min} \end{cases} \tag{12}$$

The u^{+} and u^{-} are the best estimated functions, and $|u\text{-}u^{+}|$ or $|u - u^{-}| \le \beta$, where parameter, $\beta \in [0.0, 0.1]$, is the standard deviation. The related parameters are shown in Table I.

In addition, the specific boundaries of ΔRT for screening normal subjects and subjects with PAD are parameterized with certainty factors (CFs), and the specific values of the CFs versus the ΔRTs can be presented as follows:

TABLE I
RELATED PARAMETERS FOR INFO-GAP DISTRIBUTION MODEL

$AUSPR_{nor}$	$AUSPR_{min}$	$AUSPR_{max}$	ΔRT_{min}	ΔRT_{max}	K_1^{+}	K_2^{+}
1.00	1.05	1.40	0.012 (sec)	0.100 (sec)	+ 4.16	+ 3.97
1.00	0.60	0.95	0.012 (sec)	0.100 (sec)	– 4.16	– 3.97

$$CF_1 = \begin{cases} 1, & 0 \le \Delta RT \le \Delta RT_{\min} = Mean_1 \\ \exp(-\dfrac{1}{2} \times (\dfrac{\Delta RT - Mean_1}{\sigma_1})^2), \\ & \Delta RT > Mean_1 \end{cases} \tag{13}$$

$$\Delta RT_1^{-} = \frac{-1}{K_1^{+}}(AUSPR - 1.00) \text{ or}$$

$$\Delta RT_1^{+} = \frac{1}{K_1^{+}}(AUSPR - 1.00)$$

$$CF_2 = \begin{cases} \exp(-\dfrac{1}{2} \times (\dfrac{\Delta RT - Mean_2}{\sigma_2})^2), & \Delta RT < Mean_2 \\ 1, & \Delta RT \ge Mean_2 \end{cases} \tag{14}$$

$$\Delta RT_2^{-} = \frac{-1}{K_2^{+}}(AUSPR - 0.95) \text{ or}$$

$$\Delta RT_2^{+} = \frac{1}{K_2^{+}}(AUSPR - 1.05)$$

where $Mean_1$=0.012 (s) is the average ΔRT for normal subjects, $Mean_2$=0.022 (s) is the average ΔRT for subjects with PAD, and σ_j=0.5×$Mean_j$, j=1, 2, and ΔRT>0. Hence, the payoff profiles are relevant combinations of I–G distributions and certainty factors that can be used for separating normal subjects from the subjects with PAD, and they can be represented as follows:

$$payoff^{-} = (-K_2^{+} \times \Delta RT_2^{-} + 0.95) \times CF_2(\Delta RT_2^{-}) \tag{15}$$

$$payoff_{nor} = \begin{cases} (-K_1^{-} \times \Delta RT_1^{-} + 1) \times CF_1(\Delta RT_1^{-}) \\ (+K_1^{+} \times \Delta RT_1^{+} + 1) \times CF_1(\Delta RT_1^{+}) \end{cases} \tag{16}$$

$$payoff^{+} = (+K_2^{+} \times \Delta RT_2^{+} + 1.05) \times CF_2(\Delta RT_2^{+}) \tag{17}$$

Three payoff profiles within the specific ranges are shown in Figure 5(b). For decision-making applications, payoff profiles present visualizations showing robustness for each decision as a function, which provide the critical outcomes for peripheral arterial stenosis screening. Finally, logic inference with maximum composite operations are used to generate the possible result as follows:

$$\begin{aligned} S_m &= \max_{0.6 \le AUSPR \le 1.4} (payoff^{-}, payoff_{nor}, payoff^{+}) \\ &= \max_{0.6 \le AUSPR \le 1.4} [s_1, s_2, s_3], \quad \text{Index } m = 1, 2, 3 \end{aligned} \tag{18}$$

where Index $m = 1$ or $m = 3$ is used to judge the subjects with PAD, whereas Index $m = 2$ is for normal subjects, with the inequalities $payoff_{nor} > payoff^{-} > 0$ or $payoff_{nor} > payoff^{+} > 0$ for low-degree PAD. The degree of subjects

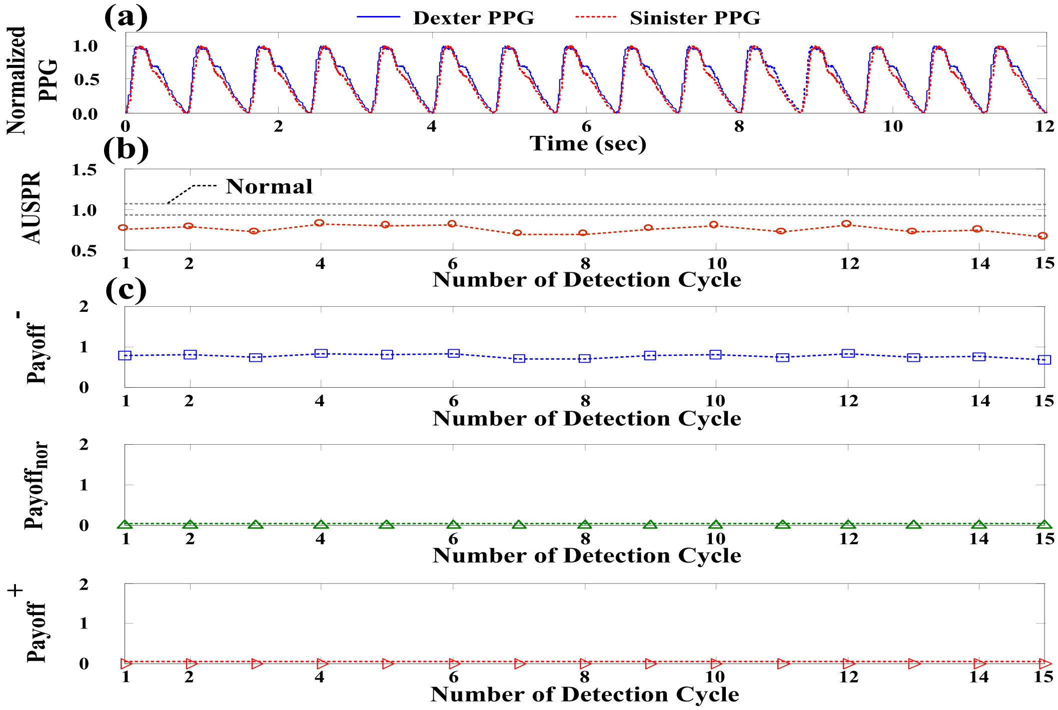

Fig. 6. Experimental results for Case Study 9#. (a) Dexter and sinister PPG measurements in time domain. (b) *AUSPR* indexes for 15 detection cycles, $AUSP_R < AUSP_L$. (c) Three payoff profiles for arterial stenose screening, $payoff^-/payoff^+$ for subjects with PAD and $payoff_{nor}$ for normal subjects.

with PAD can be divided into two levels with the following inequalities:

$$S_1 = \begin{cases} payoff^- > payoff_{nor} > 0, & for\ low\ \deg ree\ PAD \\ payoff^- > payoff_{nor} \approx 0, & for\ high\ \deg ree\ PAD \end{cases} \tag{19}$$

$$S_3 = \begin{cases} payoff^+ > payoff_{nor} > 0, & for\ low\ \deg ree\ PAD \\ payoff^+ > payoff_{nor} \approx 0, & for\ high\ \deg ree\ PAD \end{cases} \tag{20}$$

IV. Experimental Results and Discussion

This experiment was approved by the Human Research Ethics Committee and IRB. For the bilateral PPG measurements, the 35 enrolled subjects were asked to relax for about 10 min, as seen in Figure 1(c). The HD room temperature was set at 25±1 °C. Two thumb/finger clips with mounting light-emitting diodes (red-wavelength, 680 nm) and photo-detectors were placed on the dexter and sinister thumbs to detect the microcirculation blood volume pulsations by temporal analysis of scattered optical radiation. Two channel PPG signals were obtained using a data acquisition controller from the measurement system to a tablet PC [27]. A computer-assisted screening system was designed to provide physicians with a decision-making tool for signal processing, AUSPR calculations, and peripheral arterial stenosis screening. This prototype mode, which consisted of the discrete FOI computation and I-G decision-making, was implemented by using LabVIEW graphical programming software (National Instruments[TM] Corporation, Austin, Texas, USA) and MATLAB software (MathWork, Natick, Massa-chusetts, USA). Two PPG probes were placed on the sinister and dexter thumbs; the time-domain PPG measurements (approximately 12sec) are shown in Figure 6(a).

In the signal processing, an FOI with the fractional order $\alpha = 0.1$–0.4 was calculated using equations (3) to (5). The fractional order $\alpha = 0.1$ was selected to make the distinguishable features, as seen in Figures 3(c) and 3(d). Subsequently, the AUSPR indexes were calculated using equations (6) to (9). Hence, peripheral arterial stenosis screening was conducted using the I-G decision-making method. The flow chart of the overall procedure used in the proposed screening model is shown in Figure 7. For the ten HD patients, the experimental results and contrasts with bilateral timing parameters, resistive index, and fractional-order differential parameters indicated that the proposed screening model was a more efficient method, as described below.

A. Case Study: HD Patients With PAD

The experimental data from ten HD patients were employed to verify the proposed screening model. In this study, we captured at least a 1-min record for PPG measurements at the bilateral thumb sites. Figure 6(a) represents experimental measurements containing 15 time- domain PPG signals. For example, Case Study 5# has PAD and CVD, as shown in Table II. In HD patients, a case such as this would be at a higher risk of atherosclerosis, which could result in progressive dialysis vascular access stenosis. Therefore, it is important to screen early in routine HD healthcare. The proposed screening model is detailed according to the following procedure:

Step 1) obtain the sinister and dexter 15 PPG signals and find the systolic peak, then the average $AUSP_R$ and

TABLE II
NUMERICAL RESULTS IN 5 HD PATIENTS WITH PAD/CVD AND 10 HD PATIENTS WITHOUT PAD

No.	PAD	CVD	Diabetes	DOS	V_p (m/sec)	V_m (m/sec)	Resistive (Res)	ΔRT (ms)[7,10]	Ψ [14,27]	ANN [9,14]	*AUSPR* (average)	I-G Model
01	√	√	√	0.488	122	66	0.460	48.22	1.200	[0 1 0]	1.1341	1.0440 (+)
02	√	√	×	0.461	122	14	0.885	19.38	1.140	[0 1 0]	1.1357	1.1254 (+)
03	√	√	×	0.260	108	46	0.574	36.51	0.473	[0 1 0]	0.7612	0.7612 (–)
04	√	√	√	0.183	105	35	0.667	5.71	0.335	[1 0 0]	0.9326	0.2587 (–) 0.7313 (*nor*)
05	√	√	×	0.147	106	38	0.642	27.39	0.289	[1 0 0]	0.8559	0.8010 (–)
06	×	√	×	0.359	127	47	0.630	15.89	0.621	[0 1 0]	1.0357	1.0357 (*nor*)
07	×	√	×	0.347	98	5	0.948	30.68	0.522	[0 1 0]	0.9999	0.8803 (*nor*)
08	×	√	×	0.232	69	17	0.754	20.82	0.371	[1 0 0]	0.9974	0.9974 (*nor*)
09	×	√	√	0.178	99	33	0.667	31.78	0.333	[1 0 0]	0.8887	0.7425 (–)
10	×	√	×	0.098	119	44	0.630	13.15	0.252	[1 0 0]	1.1449	1.0663 (+)
11	×	×	×	0.000	85	25	0.706	4.00	0.213	[1 0 0]	0.9749	1.0000 (*nor*)
12	×	×	×	0.130	140	54	0.614	5.30	0.268	[1 0 0]	1.0755	1.0000 (*nor*)
13	×	×	×	0.147	106	38	0.642	6.60	0.290	[1 0 0]	1.0723	1.0000 (*nor*)
14	×	×	×	0.231	85	23	0.729	7.90	0.371	[0 1 0]	1.0834	1.0000 (*nor*)

Note: (1) (+) for $payoff^{+}$, (*nor*) for $payoff_{nor}$, and (–) for $payoff^{-}$; (2) ΔRT is the average bilateral rise-timing difference; (3) *AUSPR* is the average index; (4) DOS is an index that is used to evaluate the stenosis of dialysis vascular access in hemodialysis patients [28].

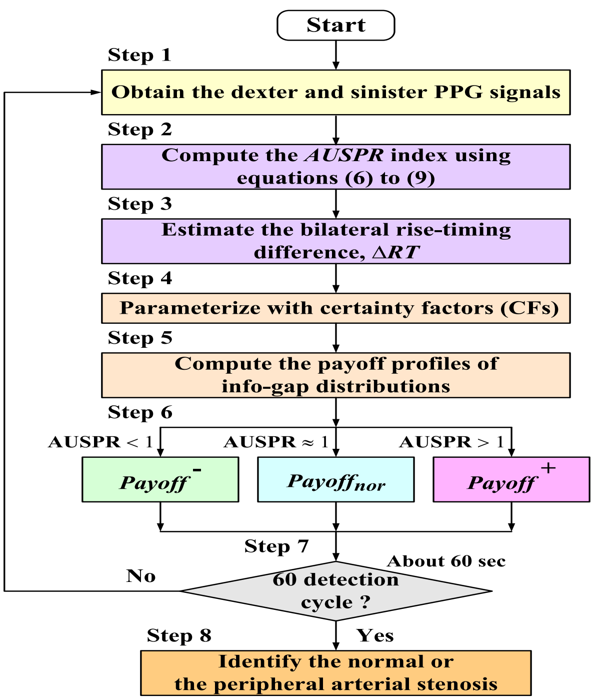

Fig. 7. The overall procedure of peripheral arterial stenosis screening.

$AUSP_L$ can be calculated using equations (3) to (5), as $[AUSP_R, AUSP_L]$=[407.51, 476.12],

Step 2) compute the *AUSPR* index using equations (6) to (9), average *AUSPR*=0.8559, and 15 detection cycles were shown in Figure 6(b),

Step 3) estimate the average bilateral rise-timing difference, $[\Delta RT_1^{-}, \Delta RT_2^{-}] = [0.0827, 0.0741]$, $\Delta RT > 0$,

Step 4) compute the certainty factors (CFs) using equations (13) and (14), $CF = [CF_1(\Delta RT_1^{-}), CF_2(\Delta RT_2^{-})] = [0, 1]$,

Step 5) compute the payoff profiles of I-G distributions as $[payoff^{-}, payoff_{nor}, payoff^{+}]$=[0.8010, 0.0000, 0.0000], average $payoff^{-}$=0.8010 and 15 detection cycles were shown in Figure 6(c),

Step 6) find the maximum value using equation (18), $S_1 = \max[payoff^{-}, payoff_{nor}, payoff^{+}]$=0.8010, indicate the possible result: "subject with PAD". For at least 15 detection cycles, the testing results confirmed Subject 9# has PAD.

Subject 9# satisfied with an inequality $payoff^{-} > payoff_{nor} \approx 0$; thus, this case study was a patient with a "high degree of PAD," as shown in equation (19). In addition, for Subject 7#, the proposed screening model indicated the following result: "healthy subject," satisfying with average $payoff_{nor} > average\ payoff^{-} = 0.2587 > 0$ (Appendix, Figure 8). This indicated that Subject #7 had low-degree PAD. In contrast with the health record, Subject 7# had PAD, CVD, and diabetes; the higher risk level indicates that the chronic diseases will be likely to induce more severe PAD and dialysis vascular access stenosis. This finding confirmed that the proposed screening model could detect peripheral arterial stenosis in its early stages in HD patients, 5 with PAD / CVD (>10 years with higher risks), 10 without PAD (<1 year or <10 years with lower risks). Data for remaining eight subjects are shown in Table II.

B. Performance Comparison and Discussion

Table II shows numerical testing comparisons from comparisons with (1) the resistive index, (2) bilateral timing parameters [7], [10], and (3) the hybrid method, which consists of the fractional-order dynamic error and an artificial neural network (ANN) [9], [14], [27]. Acoustic measurements, such as Doppler ultrasonic flowmeter measurements (using A-scan mode at a frequency 7.5-MHz), are non-invasive techniques for the assessment of vascular stenosis in blood vessels with a

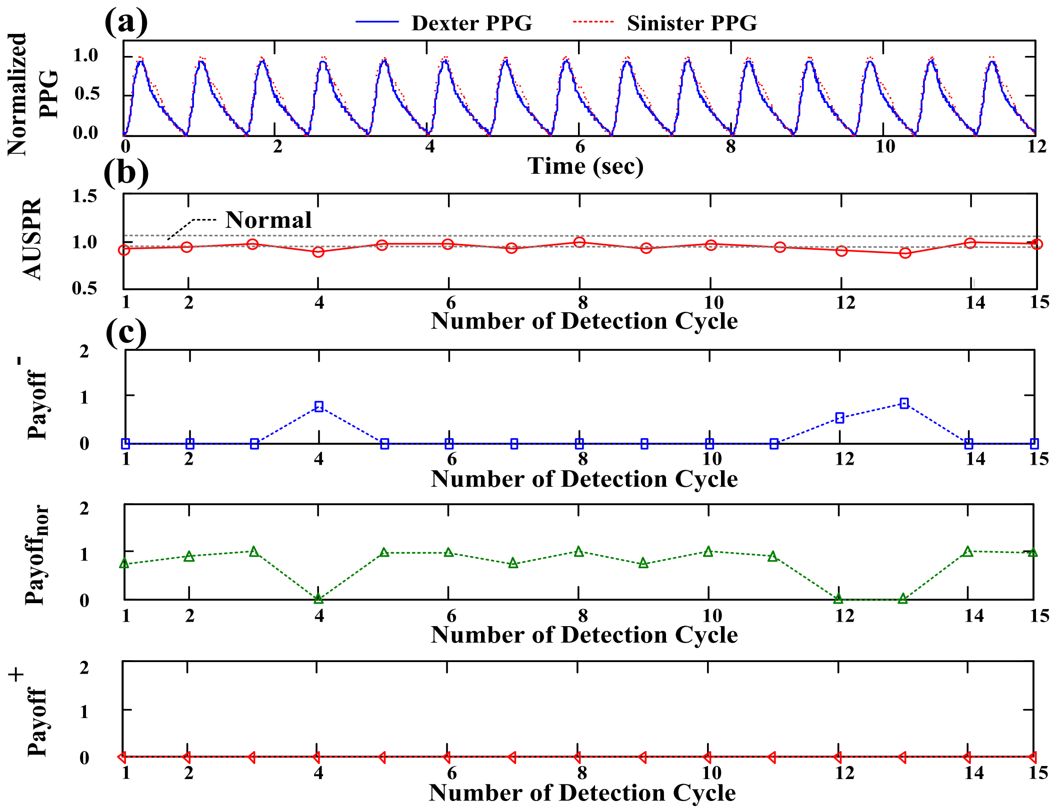

Fig. 8. Experimental results for Case Study 7#. (a) Dexter and sinister PPG measurements in time domain, (b) *AUSPR* indexes for 15 detection cycles, $AUSP_R < AUSP_L$, (c) Three payoff profiles for arterial stenose screening, $payoff^-/payoff^+$ for subjects with PAD and $payoff_{nor}$ for normal subjects.

diameter of 5–12 mm. This technique can be used to measure velocity profiles and identify peak-systolic velocity (PSV, V_p) and peak-diastolic velocity (PDV, V_m), as shown in Table II. The resistive index can be calculated using PSV and PDV (with the formulation Res $= (V_p - V_m)/V_p$) [27], [29]. This index is an indicator of vessel resistance, and the normal range for the common vessel is 0.50–0.65. Higher values indicate the risk level of CVDs or vascular stenosis. In the health records, subjects had CVD or PAD, and some subjects with type 2 diabetes mellitus had lower-limb PAD. However, at the measurement site, an undetected narrowed access and multi-site stenosis, the Doppler angle, and the Doppler shift frequency selection could all affect the efficiency of the acoustic technique, e.g., as with Subject 1#.

Bilateral timing differences, such as RTs and PTTs, have primarily been used as screening parameters to prevent progression to significant PAD in patients with CVD and diabetes mellitus. Previous studies [7], [10]–[12] showed that when PAD gradually deteriorates, the transit time and shape of bilateral PPG signals becomes asymmetric. Time delays increase and the PPG pulses change in terms of amplitude and shape. In both the lower and upper limbs, these timing parameters differed more significantly in patients with PAD. The timing parameters were prone to disturbance by heart-rate variations and quantification errors for parameter extraction in time domains, as shown in Table II. In addition, patients with PAD could develop various morphologic changes in their systolic rising edge. Considering this, the fractional-order dynamic error was also employed to extract the dexter-to-sinister differences involving rising time and amplitudes in the systolic rising edge. The dynamic errors, Ψ, increased with the severity of dialysis vascular access stenosis [14], [27]. An ANN was proposed for categorizing the degree of stenosis, as vector=[*normal*, *lower degree*, *higher degree*] was encoded as binary values with value "1" or "0" for PAD screening in Table II. The traditional least-square algorithm or gradient descent method [9], [14] was used to determine the ANN's parameters. However, determination of the multi-layer network's structure and the update of parameters with iterative computations were two issues that limited the mechanism's inclusion in a portable embedded system.

A discretization FOI with the fractional order $\alpha = 0.1$–0.2 was used to extract features in the signal pressing stage. This FOI employed non-integer numbers to deal with signals, and the summation was computed with the ratios of the Gamma function using equations (3) and (4), which incorporated the number of sampling data points and the fractional order parameters [19]. Thus, the distinguishable features could be extracted in time and frequency domain analysis, e.g., for variations of RT, settling-time, peak-time, and overshoot edges. In addition, the I-G decision-making method can be used to overcome the complexity of an adjustable mechanism. It flexibly divided the margins to separate normal subjects from subjects with PAD. The possible result could also be divided into very likely (VL) and likely (L), including the following subdivisions: (1) normal subjects (VL) or normal subjects with low-degree PAD (L) and (2) subjects with high- degree PAD (VL) or low-degree PAD (L), as seen in Figure 5(b).

V. CONCLUSION

In Taiwan, ESRD and diabetes are common irreversible and progressive chronic diseases within the five ranks. HD patients with CVD and diabetes mellitus are often affected by complications such as PAD and HD vascular access. Hence, in this study, we proposed an assistant detection model consisting of FOI and inference methods for the screening of peripheral arterial stenosis. A discretization FOI with the fractional order $\alpha = 0.1$–0.2 can be designed by using power series expansion, and it can be implemented in real time signal applications. An I-G decision-making method was used as a flexible inference mechanism to identify the critical outcomes, which included two categories: (1) normal subjects without PAD or normal subjects with low-degree PAD and (2) subjects with high-degree PAD or low-degree PAD.

For ten randomly selected HD patients, the experimental results showed that the proposed model was highly accurate, and a positive predictability of $>80\%$ was obtained when quantifying the performance of the proposed model with or without a slightly noisy background. Comparisons of the proposed method with (1) the resistive index, (2) bilateral timing parameters, and (3) the hybrid method showed that it was more efficient in peripheral arterial stenosis screening. In addition, the inference mechanism of the method required no iterative computation to update any parameter. Thus, the proposed screening model could easily be implemented in an embedded system. In an embedded system, its technique could be applied to design a detector, e.g., using a built-in digital signal processor to support signal preprocessing or implementing the feature extraction and manner of inference in a programmable processor.

APPENDIX

Case Study 7# has CVD, PAD, and diabetes. Figure 8(a) shows the raw data of dexter and sinister PPG signals in time domain. Figures 8(b) and 8(c) are the experimental results.

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